

Fundamental of Measurement and Sensors

Measurement

- Measurement is the process by which one can convert physical parameter (what we have to measure) to meaningful numbers.
- Measurement is process of quantifying a property of an object or phenomenon using a standard units
- Gives the Answer

How long
How much,
How heavy
How hot e.t.c

In order that the results of the measurement are meaningful there are two basic requirements:

- The standard used for comparison purposes must be accurately defined and should be commonly accepted.
- The apparatus used and the Method adopted must be provable

Key component of Measurement

- ① What to measure? → ~~Extr~~ physical quantity being measured (length, time etc)
- ② unit of measurement (meter, second, Ampere)
- ③ ~~The~~ Instrument → measuring instrument (Ruler, thermometer)

Significance or Importance of Measurement

- Daily life $\rightarrow$ Buying, selling, time, cooking, clothing
- Medicine $\rightarrow$ Dose, Diagnostics, Test
- Science & Technology $\rightarrow$ To make accurate experiments & discoveries
- Engineering $\rightarrow$ Design equipment & process

$\rightarrow$ Measurement gives a ~~star~~ satisfaction when results are expressed in numbers.

$\rightarrow$ Advancement of science & Technology is dependent on parallel progress in measurement technique.

$\rightarrow$ Two major functions of all branches of engineering:

- (i) Design of equipment & processes
- (ii) proper operation and maintenance of equipment and processes.

Both above functions require measurements.

$\rightarrow$ Measurement plays a significant role in achieving goals and objectives of engineering because of the feedback information supplied by them.

Classification of Instruments

① Broad classification

- (i) Absolute instruments or primary instruments
- (ii) Secondary instrument

(i) Absolute instrument (seldom used except standard institution)

→ A type of instruments used mainly for standardizing instruments in laboratories.

→ Examples:- (a) Rayleigh current balance
(b) Tangent galvanometer

→ These instruments give the magnitude of the quantity under measurement in terms of physical constants of the instrument.

(ii) Secondary instruments (most commonly used)

→ These instruments are so constructed that the quantity being measured can only be measured by observing the output indicated by the instrument.

→ These instruments are calibrated by comparison with absolute instrument or another secondary instruments which has already calibrated against an absolute instruments.

→ Examples:- (a) voltmeter
(b) Glass thermometer
(c) pressure gauge

③ Based upon the way they present the result of measurements.

Instrument can be classified into two major categories: (i) Deflection type

(ii) Null detector type

(i) Deflection type instrument

→ More suited for measurement under dynamic condition than null type instruments

→ In this type of instruments, the deflection of the instruments provides a basis for determining the quantity measurement. The measured quantity produce some physical effect with deflection or produces a mechanical displacement of the moving system of the instrument.

→ Examples: (a) PMMC
(b) MI

(ii) Null type instrument

→ High accuracy as compared to deflecting type.

→ Highly sensitive as compared to with deflection type.

→ In this type of instruments a zero or null indication leads to determination of the magnitude of measured quantity. The null condition is dependent upon some other known quantities.

→ Examples: (a) LVDT
(b) potentiometer

③ Mode of operation

Secondary Instrument work in two modes

so, it can be classified as (1) Analog Instrument
(2) Digital Instrument

(1) Analog Instruments

→ An analog instrument is one in which the output or display is a continuous function of time and bears a constant relation to its output input.

→ Examples ÷

- (a) voltmeter
- (b) wattmeter
- (c) energymeter
- (d) x-y plotter

Classification of analog instruments ÷

(i) on the basis of the quantity they measure.

- (a) Ammeter → An instrument meant for measurement of current is classified as an ammeter.
- (b) voltmeter → Measure voltage
- (c) wattmeter
- (d) Energymeter
- (e) power factor meter
- (f) Frequency meter

of Analog Instruments

(ii) Classification^{of Analog Instruments} on the basis of their nature of operations:-

- (a) Indicating Instruments
- (b) Recording Instruments
- (c) Integrating Instruments

(a) Indicating Instruments

- Instruments which indicate the magnitude of quantity being measured. They generally make use of a dial and a pointer for this purpose.
- Damping system decides the time of response of an indicating instrument.
- Examples:- Voltmeter, Ammeter, Wattmeters etc.

Ⓐ Analog indicating instruments may be divided into two groups:-

- Electromechanical Instruments
- Electronic Instruments

(b) Recording Instruments

- Instruments which give a continuous record of the quantity being measured over a specified period of time.
- Examples:- Aneroid barometer, Null balance, x-y plotter

(c) Integrating Instruments

- Instruments which give totalized event over a specified period of time. The summation which they give is the product of time and an electrical quantity.
- Energy meter, A-h meter

(iii) Classification of Analog instrument on the basis of Method used for comparing the unknown quantity with the unit of measurement.

→ The two categories on the above basis :-

(a) Direct measuring instruments

(b) Comparison instruments

(a) Direct measuring instruments

→ These instruments convert the energy of the ~~measured~~ measurand directly into energy that actuates the instrument and the value of the unknown quantity is measured or displayed or recorded directly.

→ Examples :- Ammeters
Wattmeters
Voltmeters
Energy meters

(b) Comparison instruments

→ These instruments measure the unknown quantity by comparison with a standard

→ Examples :- DC bridges
AC bridges

→ comparison type instruments are used in cases where higher accuracy of measurement is required.

(iv) Classification of secondary analog instruments according to the principle of operation they utilize

→ The effect they utilize are :-

- (a) Magnetic effect
- (b) Heating effect
- (c) Electrostatic effect
- (d) Electromagnetic effect
- (e) Hall effect

(a) Magnetic effect

Examples :- Ammeter, voltmeters, wattmeters
Integrating meters.

(b) Heating effect

Examples :- Ammeter & Voltmeters

(c) Electrostatic effect

Examples - Voltmeters

(d) Electromagnetic effect

Examples - Ac Ammeter, Ac voltmeters

(e) Hall effect

Examples :- Flux meters, Poynting vector wattmeters

(2) Digital Instruments

→ An electronic instruments that converts a physical quantity into digital signal and display the measured numerically on a screen.

→ Examples: DMM (Digital multimeters)

Digital thermometers

Digital tachometers

Digital weighing scale

① Electrical Instrument may also be classified on the basis of the kind of current that can be measured by them.

→ Electrical Instrument may be classified as instruments for:-

(a) Direct current (d.c) eg. PMMC

(b) Alternating current (a.c) eg - energy meters

(c) Both d.c and a.c / universal instruments

eg → MI
EMMC
Thermocouple

operating forces

(6)

→ Those types of forces are needed for the satisfactory operation of any indicating instrument these are:-

(1) Deflecting forces

(2) Controlling forces

(3) Damping forces

(1) Deflecting forces

→ The deflecting or operating force is required for moving the pointer from its zero position.

→ The system producing the deflecting force is called deflecting system or moving system.

→ The deflecting force can be produced by utilizing heating effect, magnetic effect, electrostatic effect, induction effect etc

→ Deflecting system of an instrument converts the electric current or potential into mechanical force called deflecting force

(2) Controlling force

→ This force is required in an indicating instrument in order that the current produces deflection of the pointer proportional to its magnitude.

→ The system producing a controlling force is called a controlling system.

function of controlling system are:-

- To produce a force equal and opposite to the deflecting force at the final steady position of pointer in order to make the deflection of the pointer definite for a particular magnitude of current.
- To bring the moving system back to zero.

(3) Damping force

→ When a deflecting force is applied to the moving system, it deflects and it should come to rest at a position where deflecting force is balanced by controlling force. T_d & T_c produce by system which have inertia and therefore moving system can't immediately settle at its final position but overshoot & oscillate ahead of it.

Constructional details of Indicating Instrument ⑦

(i) Moving system

- moving parts should be light.
- Frictional force should be minimum.

(ii) supporting the moving element

Type of supports

- (a) suspension
- (b) Taut suspension
- (c) pivot and jewel bearing
- (d) balancing

Control systems

There are two types of control system which are used for mounted system are:-

(i) Gravity control

→ In this type of control, a small weight is placed on an arm attached to the moving system. The position of this weight is adjustable. This weight produces a controlling torque due to gravity

→ Vertical mounted system

→ $T_c \propto \sin \theta$

(1) Spring control

→ A hair spring attached to the moving system exert a controlling torque.

Spring

↳ should be non-magnetic

↳ should be proof from mechanical fatigue

→ vertical and horizontal

→ $T_c \propto \theta$

Damping system

The methods for producing damping torque are:

(i) Air friction damping → MI, EMMC, EDM

(ii) Fluid friction damping → Electrostatic voltmeter

(iii) Eddy current damping → PMMC

(iv) Electromagnetic damping → Galvanometer

(i) Air friction damping

concept and terminology of transducers

→ A transducer is a device which, when actuated transform energy from one form to another.

In the context of measurement and instrumentation, transducer typically converts a physical / non-electrical quantity (Temp, pressure, light e.t.c) into an electrical signal. This electrical signal can be measured, Record or used to control a system

Eg → (i) A microphone is a transducer that convert sound (mechanical energy) into electrical signal.

(ii) A thermocouple convert heat (thermal energy) into an electrical voltage.

* ~~1~~ # Terminology Related to transducer

* Another name of a transducer is pick up.

* Actuator → Converts electrical signals to physical action (opposite of sensor)

It is the O/P element of a control system and used to move or control a mechanism or system.

In simple terms, while a sensor detects changes in the environment, an actuator causes changes in the environment.

Eg → Electric motor
Solenoid
Hydraulic brake

* sensor $\rightarrow$ A part of the transducer that detects the physical quantity and respond to it.

$\rightarrow$ A sensor detects the changes in the environment or a system and provide input to a controller or processor.

Eg $\rightarrow$ (a) Temperature sensor $\rightarrow$ Thermocouple
Thermistor
 $\rightarrow$ used in AC, thermostats

(b) proximity sensor $\rightarrow$ infrared sensor
ultrasonic sensor
 $\rightarrow$ used in mobile phone (screen on/off)
Automatic doors

(c) Light sensor $\rightarrow$ LDR, photodiode
 $\rightarrow$ used in street light (Automatic on/off)
smartphone (brightness Adjustment)

classification of transducer

① on the basis of transduction form used

(1) Resistive transducer - passive

→ change in resistance due to variation in input quantity.

Eg • strain gauge → measure force or strain
• Thermistor → measure temperature

(2) Inductive transducer - passive

→ change in inductance due to changes in the physical parameters.

Eg → • LVDT → measure linear displacement
• proximity sensor → detect nearby metallic object

(3) Capacitive transducer - passive

→ change in capacitance due to change in distance b/w plates, area of plates or dielectric material

Eg → Dielectric gauge → liquid level measurement or thickness

Variable capacitor pressure gauge → measure displacement of pressure

(4) voltage and current

→ change in potential difference or current detection or dirn due to change in physical parameters.

Eg → Hall effect pickup → Measure magnetic flux current

Ionization chamber → Measure particle counting, radiation

(5) self generating transducers

→ NO external power required.

- Thermocouple & thermopile → Measure Temp, Heat flow

- Piezoelectric pickup → measure sound vibrations, pressure changes

- Photovoltaic → used in light meter solar cell

② As primary and secondary transducers

(1) Primary transducer → A primary transducer is the first element that sense or detect the physically quantity directly and converts it into another form of energy (not necessarily electrical)

Eg → Bourdon tube → converts pressure into displacement

Thermocouple → convert heat into voltage

(2) Secondary transducers

→ A secondary transducer take the output from the primary transducer and converts it into electrical signal that can be measured or recorded.

Eg → LVDT →

Strain gauge →

Potentiometer

(C) As Active and passive transducers

(1) Active transducers

→ Active transducer are those which do not require an auxiliary power source to produce their output. They are also known as self generating type since they develop their own voltage or current output.

Eg → Thermocouple

Photovoltaic cell

Tachogenerator

(2) passive transducer

→ passive transducers derive the power required for transduction from an auxiliary power source. They also derive part of the power required for conversion from the physical quantity under measurement. They are also known as externally powered transducer.

Eg - Resistive, inductive

① As Analog and Digital transducers

The transducers can be classified on the basis of the output which may be a continuous function of time or the output may be in discrete steps

(I) Analog transducer

→ These transducers convert the input quantity into an analog output which is a continuous function of time.

Eg → strain gauge

LVDT

Thermocouple etc

(II) Digital transducers

→ These transducers convert the input quantity into an electrical output which is in the form of pulses.

PMMC Instrument (Permanent magnet moving coil Instrument)

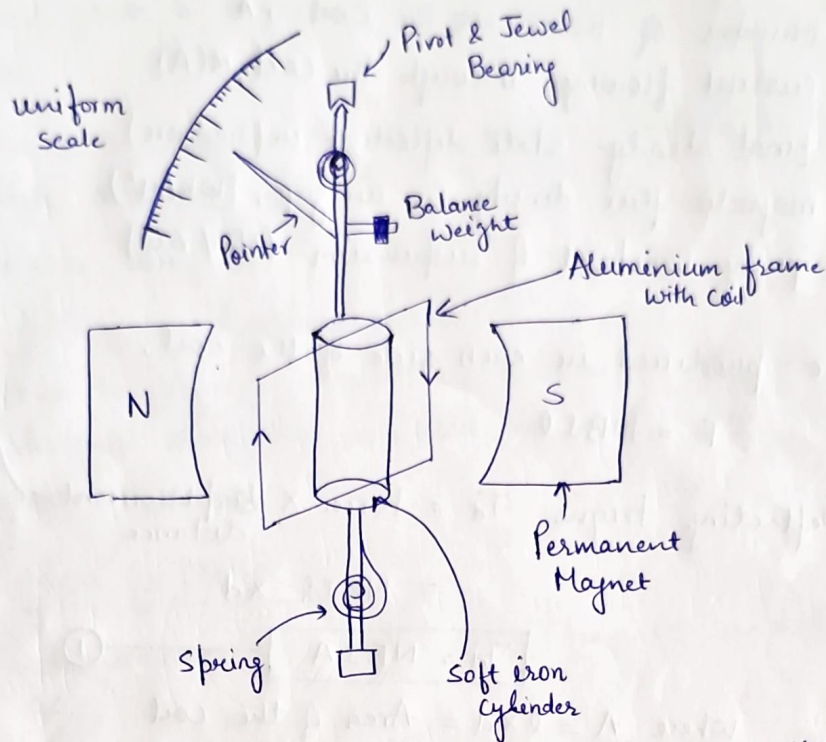


Fig shows the schematic view of PMMC indicating its different parts. A light rectangular coil wound over an aluminium frame is pivoted within the air gaps between the magnet poles N & S and a cylindrical soft iron core.

The spindle which supports the aluminium frame also carries a pointer. The aluminium frame supports the coil and provides eddy current damping. The controlling torque is provided by two phosphor bronze hair springs coiled in opposite directions, which also serve as leads for current in the coil.

When current passes through the coil, a deflecting torque is produced due to interaction between permanent flux and flux produced by moving coil.

let

l, d = length of vertical and horizontal side of coil (m)

N = Number of turns in the coil, *

I = current flowing through the coil, (A)

θ = final steady state deflection in (radian)

B = magnetic flux density in air gap, (wb/m^2)

K = Spring Constant of suspension (Nm/rad)

The force produced in each side of the coil,

$$F = NBIL$$

$\therefore$ Deflecting torque $T_d = \text{Force} \times \text{displacement distance}$

$$= NBIL \times d$$

$$\boxed{T_d = NBAI} \quad \text{--- (1)}$$

where $A = l \times d = \text{Area of the coil}$

For this instrument, N, B , and A are constant.

$$T_d = GI$$

where $G = NBA = \text{constant}$

The expression for controlling torque in case of spring controlled instrument is given by

$$T_c = K\theta$$

$\therefore$ For steady state deflection

$$T_c = T_d$$

$$\Rightarrow K\theta = GI$$

$$\theta \propto I$$

Since θ is proportional to I , therefore scale is uniformly divided and can be used for the measurement of DC quantity only.

Advantages of PMMC

- 1) Very small power is required for its operation. Hence, power consumption is very low.
- 2) The torque to weight ratio of moving system is high which gives high accuracy.
- 3) The scale is uniform.
- 4) It is free from hysteresis error.
- 5) Error due to stray magnetic field is negligible.
- 6) Eddy current damping is used without any extra damping mechanism.

Disadvantages of PMMC

- 1) These instruments are useful only for dc and cannot be used for ac.
- 2) The cost of instrument is higher than the moving iron instrument.
- 3) Thermoelectric emf may cause errors when it is used with shunts.
- 4) With time the strength of permanent magnet reduces, causing error in reading.

Moving Iron Instrument (MI)

Moving iron instruments are used to measure current and voltage with an accuracy.

Working Principle -

A plate or vane of soft iron forms the moving element of the system. This iron vane is so situated that it can move in a magnetic field produced by a stationary coil. The coil is excited by current or voltage under measurement, which makes it an electromagnet and the iron vane moves in such a way so as to increase the inductance of the coil.

flux of electromagnet. Thus the force produced is always in such a way so as to increase the inductance of the coil.

Equation of Torque—

Suppose initial current is I , the instrument inductance L and the deflection θ . If the current increased by dI then the deflection changes by $d\theta$ and the inductance by dL . In order to affect an increment dI in the current there must be an increase in the applied voltage given by,

$$e = \frac{d}{dt}(LI) = I \frac{dL}{dt} + L \frac{dI}{dt}$$

Total electrical energy supplied

$$eI dt = I^2 dL + LI dI$$

Change in stored energy is given by

$$\Delta E = \frac{1}{2} (I + dI)^2 (L + dL) - \frac{1}{2} I^2 L$$

$$\Rightarrow \Delta E = \frac{1}{2} \{ I^2 + 2IdI + (dI)^2 \} (L + dL) - \frac{1}{2} I^2 L$$

$$\Delta E = \cancel{\frac{1}{2} I^2 L} + \frac{1}{2} I^2 dL + ILdI + IdI dL - \cancel{\frac{1}{2} I^2 L}$$

Neglecting all smaller terms of ΔE

$$\Delta E = \frac{1}{2} I^2 dL + ILdI$$

let T_d be the deflecting torque

$$\therefore \text{Mechanical work done} = T_d d\theta$$

From the principle of Conservation of energy

$$\text{Electrical Energy Supplied} = \text{Increase in Stored energy} + \text{Mechanical work done}$$

$$\Rightarrow I^2 dL + IL dI = \frac{1}{2} I^2 dL + IL dI + T_d d\theta$$

$$\Rightarrow T_d d\theta = \frac{1}{2} I^2 dL$$

$$\Rightarrow \boxed{T_d = \frac{1}{2} I^2 \frac{dL}{d\theta}} \quad \text{--- (1)}$$

$$\text{Controlling torque, } T_c = K\theta \quad \text{--- (2)}$$

At equilibrium, $T_c = T_d$

$$\Rightarrow K\theta = \frac{1}{2} I^2 \frac{dL}{d\theta}$$

$$\theta \propto I^2 \quad \& \quad T_d \propto I^2$$

Thus, the deflecting torque T_d is unidirectional whatever may be the polarity of the current. Hence, moving iron instrument can be used for measurement of both d.c. & a.c. quantity.

Classification of moving Iron instruments

(i) Attraction type

(ii) Moving type

(1) Attraction type MI

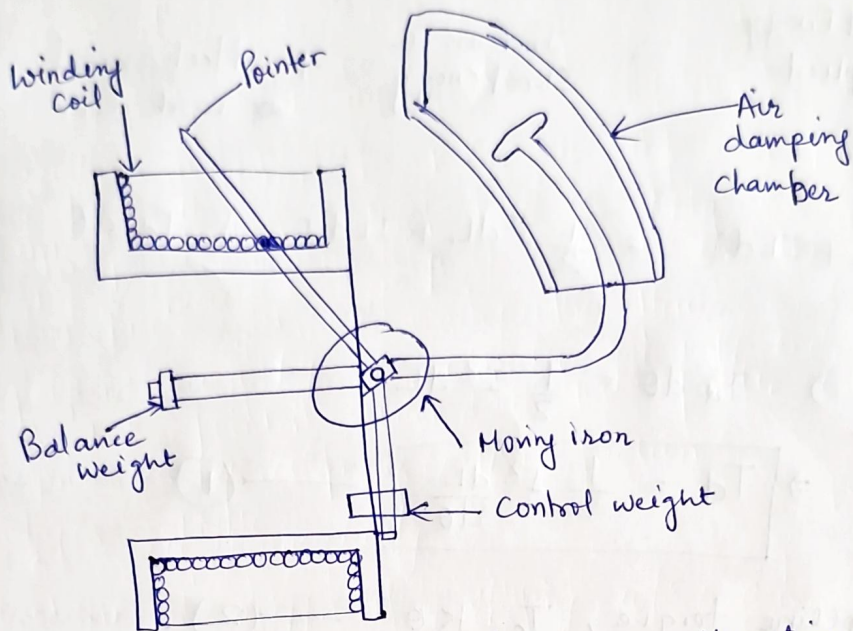


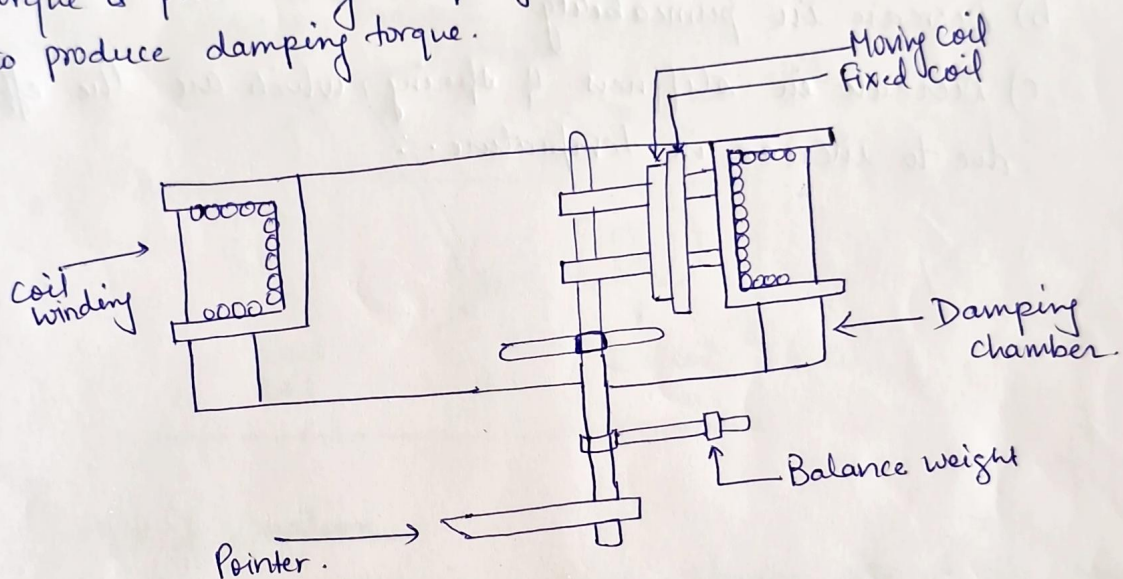
Fig shows the constructional detail of an attraction type MI instrument. The coil is ~~not~~ flat and has a narrow slot like opening. The moving iron is a flat disc or a sector ~~at~~ eccentrically mounted. When the current flows through the coil, a magnetic field is produced and the moving iron moves from the weaker field outside the coil to the stronger field inside it or in other words, the moving iron is attracted in. Damping is provided by air friction with the help of a light aluminium piston attached to the moving system.

(2) Repulsion type MI instrument

Fig shows the constructional details of an Repulsion type moving iron instrument. Insulated Copper wire is wound around a bobbin in the form of a coil. The size of wire and number of turns depends upon the ampere turns required and the magnitude of current to be measured. Inside the space ~~the~~ of bobbin (coil), two pieces of soft iron are

placed face-to-face. One is fixed on the wall of bobbin which is called as fixed iron and the other is fixed to spindle which is free to move and called moving iron.

When a current is passed through the coil, a magnetic field is produced within the hollow space. The two iron pieces are within the field. They are magnetised in the same direction of polarity, irrespective of direction of current in the coil. Hence, the two iron pieces will repel each other. Hence, a deflecting torque is produced and moving iron is deflected. The deflection is indicated by pointer attached to the spindle. Controlling torque is provided by a spring. Air friction damping is adopted to produce damping torque.



Advantages of MI instrument:-

- i) it can be used on both AC and DC system.
- (ii) Have high operating torque.
- (iii) Suitable for low frequency & high power circuits.
- (iv) It can be built at less cost than other types.
- (v) Instruments are robust owing to simple construction.

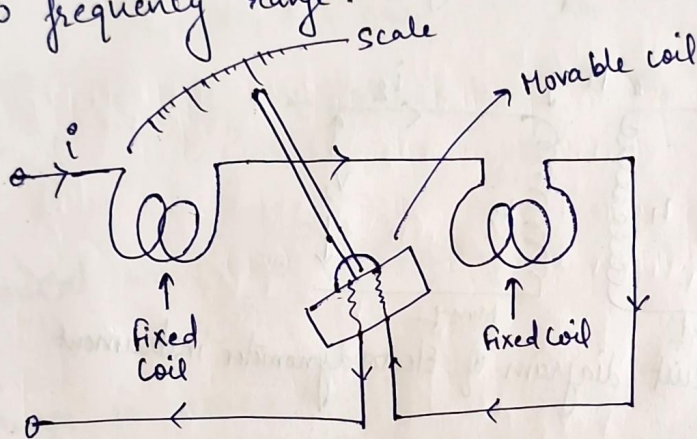
~~Disadvantages~~

Disadvantages of MI

- 1) Error is caused due to hysteresis in the iron.
- 2) The readings are affected due to stray fields.
- 3) The scale is not uniform because the deflection $\theta \propto I^2$
- 4) The instrument readings are true for only one frequency for which it has been designed.
- 5) The reading shown by the meter get affected by —
 - a) Increase in resistance coil
 - b) Decrease in permeability
 - c) Decrease in stiffness of spring, which are the effects due to increase in temperature.

Electrodynamometer type instruments

- Electrodynamometer type instruments use the characteristics of a transfer type instrument i.e., it can be calibrated with a d.c. source and then used without modification to measure a.c.
- It has same accuracy for both ac and dc unlike moving iron instrument.
- These instruments are used as ac voltmeters and ammeters both in the range of power frequencies and lower part of the audio frequency range.



Principle of operation

When the ac voltage is applied a torque acts on the movable coil in one direction during one half of the cycle and an equal torque in the opposite direction during the other half of the cycle.

Construction

An electrodynamometer type instrument consists of two types of coils namely fixed coil and a moving coil. Fixed coil is used to produce required field and is divided into two sections to give a more uniform field near the centre and to

allow passage of the instrument shaft. It consists of one moving coil with the pointer is connected which shows the value of the measured value of the scale.

Equation of Torque

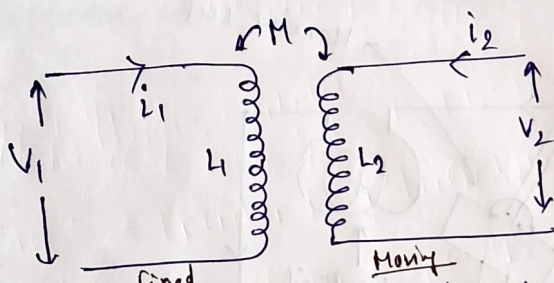
let i_1 = current in the field coil at time t

i_2 = current in moving coil at time t

M = mutual inductance b/w two coils

L_1 = Inductance of fixed coil

L_2 = inductance of moving coil



Circuit diagram of Electrodynamometer instrument

Flux linkage with fixed coil,

$$\Psi_1 = (L_1 i_1 + M i_2)$$

Flux linkage with moving coil

$$\Psi_2 = (L_2 i_2 + M i_1)$$

Vtg induced in fixed coil is

$$V_1 = \frac{d\Psi_1}{dt} = \frac{d}{dt}(L_1 i_1 + M i_2) \quad \text{--- (1)}$$

Voltage induced in moving coil is

$$V_2 = \frac{d\psi_2}{dt} = \frac{d}{dt}(L_2 i_2 + M i_1) \quad \text{--- (2)}$$

$$\text{Total i/p energy in the two coils} = V_1 i_1 dt + V_2 i_2 dt \quad \text{--- (3)} \\ (\because E = P \times t)$$

Substituting eqⁿ (1) & (2) in eqⁿ (3)

$$\begin{aligned} \text{i/p energy} &= i_1 d(L_1 i_1 + M i_2) + i_2 d(L_2 i_2 + M i_1) \\ &= i_1 L_1 di_1 + i_1^2 dL_1 + i_1 M di_2 + i_1 i_2 dM \\ &\quad + i_2 L_2 di_2 + i_2^2 dL_2 + i_2 i_1 dM + i_2 M di_1 \end{aligned} \quad \text{--- (4)}$$

$$\text{Mechanical work done,} = T d\theta \quad \text{--- (5)}$$

$\therefore$ Change in stored energy

$$= d \left[\frac{1}{2} i_1^2 L_1 + \frac{1}{2} i_2^2 L_2 + i_1 i_2 M \right]$$

$$\begin{aligned} &= L_1 i_1 di_1 + \frac{i_1^2}{2} dL_1 + L_2 i_2 di_2 + \frac{i_2^2}{2} dL_2 \\ &\quad + i_1 i_2 dM + i_1 M di_2 + i_2 M di_1 \end{aligned} \quad \text{--- (6)}$$

Now Total i/p electrical energy = change in stored energy + Mechanical work done

$$\frac{1}{2} L I^2$$

$$\begin{aligned}
 i_1 L_1 di_1 + i_1^2 dL_1 + M i_1 di_2 + i_1 i_2 dM + i_2 L_2 di_2 + i_2^2 dL_2 + M i_2 di_1 \\
 + i_1 i_2 dM = L_1 i_1 di_1 + \frac{i_1^2}{2} dL_1 + L_2 i_2 di_2 + \frac{i_2^2}{2} dL_2 \\
 + i_1 i_2 dM + i_1 M di_2 + i_2 M di_1 \\
 + \tau_d d\theta
 \end{aligned}$$

$$\Rightarrow \frac{1}{2} i_1^2 dL_1 + i_1 i_2 dM + \frac{1}{2} i_2^2 dL_2 = \tau_d d\theta$$

Since $L_1, L_2 = \text{constant}$ & $\therefore dL_1 \text{ \& } dL_2 = 0$

$$\tau_d d\theta = i_1 i_2 dM$$

$$\boxed{\tau_d = i_1 i_2 \frac{dM}{d\theta}} \rightarrow \text{Reflecting torque,}$$

Controlling torque $\Rightarrow \tau_c = k\theta$

At balance Condition,

$$\tau_c = \tau_d$$

$$k\theta = i_1 i_2 \frac{dM}{d\theta}$$

$$\theta = \frac{i_1 i_2}{k} \frac{dM}{d\theta}$$

$$\boxed{\theta \propto i_1 i_2}$$

Advantages of Electrodynamometer

- i) It is free from hysteresis and eddy current errors, ~~but is used only as~~ (coils are air cored)
- (ii) Can be used for both ac as well as dc measurements.
- (iii) Accuracy is very high.

Disadvantages of Electrodynamometer

- (i) low torque to weight ratio hence have low sensitivity.
- (ii) The scale is non-uniform, as deflecting torque is proportional to square of voltage or current.
- (iii) Frictional losses are high.
- (iv) Highly expensive as compared other instrument for ammeter, voltmeter, therefore used for precision measurement.
- (v) High power consumption as compared to PMMC instrument.

Errors and it's type in measuring instruments

Errors

Errors in measuring instruments are deviations b/w the measured value and the true value of the quantity being measured.

These errors can arise from various sources and are generally classified into following types:-

Types of errors

(1) Gross errors

(2) Systematic errors

- (a) Instrumental error
- (b) Environmental Errors
- (c) observational Errors

(3) Random errors

(1) Gross errors

This class of errors mainly covers human mistakes in reading instruments and recording and calculating measurement results.

The responsibility of the mistake normally lies with the experimenter.

may grossly misread the scale

Eg → Due to oversight read the temp 31.5°C while actual reading may be 21.5°C

→ may transpose the reading while recording read 25.8°C & record 28.5°C

Human beings are involved, so some gross errors will definitely be committed. Although complete elimination of gross errors is probably impossible, one should try to anticipate and correct them.

Gross errors may be of any amount and therefore their mathematical analysis is impossible. They can be avoided by adopting two means:-

- (1) Great care should be taken in reading and recording the data.
- (ii) Two, three or even more reading should be taken for the quantity under measurement.

(2) Systematic Errors

(i) Instrumental Errors

→ These errors arise in instrument due to three main reasons:-

- (a) Due to inherent shortcoming in the instrument.
- (b) Due to misuse of the instruments.
- (c) Due to leading effects of instruments.

To avoid ~~the~~ reduce above three instrumental error some precaution/measure should be taken which is given below:-

- (1) For Inherent shortcoming of the instrument
The procedure of measurement must be carefully planned, correction factor should be applied after determining the instrumental errors and instrument may be re-calibrated carefully.

④ In case it is suspected that external magnetic or electrostatic fields can affect the readings of the instruments, magnetic or electrostatic shields may be provided.

⑤ Apply computed corrections.

(III) Observational errors

There are many sources of observational errors.

- parallax error will be incurred unless the line of vision of observer is exactly above the pointer. To minimize parallax errors, highly accurate meters are provided with mirrored scales.

Since the parallax errors arise on account of pointer and the scale not being in the same plane.

- There are human factors involved in measurement. The sensing capabilities of individual observer affect the accuracy of measurement. No two person observe the same situation in exactly the same way where small details are concerned. For examples there are observational errors involving timing of an event. One observer may tend to anticipate the signal & read too soon. Different experimenters may produce different result especially when sound & light measurement are involved since no two observer possess the same physical response.

Modern electrical instruments have digital display of output which completely eliminates the errors on account of human observational or sensing powers as o/p is in the form of digit.

(i) For ~~to~~ avoid the errors due to misuse of instruments, it should be taken care of correct handling and uses, training & supervision and error documentation & correction.

(ii) To avoid ^{error due to} loading effect in instruments use high i_p impedance instruments, use less low i_p impedance sources, proper selection of instrument range and minimize measuring time.

(ii) Environmental Errors

These errors due to conditions external to the measuring device including condn in the area surrounding the instruments. These may be effects of temperature, pressure, dust, humidity, vibration or of external magnetic or electrostatic ~~of~~ fields.

The corrective measures employed to eliminate or to reduce these undesirable effects are:

- (a) Temperature can be kept constant by keeping the equipment
- (b) Variation in resistance with temp can be minimized by using resistance material which have a very low resistance temperature co-efficient.
- (c) The effect of humidity, dust etc can be entirely eliminated by hermetically sealing the equipment.

(3) Random Errors

→ It is found that experimental results show variation from one reading to another, even after all systematic errors have been accounted for. These errors are due to multitude of small factors which change or fluctuate from one measurement to another and are due solely to chance. The quantity being measured is affected by many happenings throughout the universe. we are aware of some factors influencing the measurement but about the rest we are unaware.

The happenings or disturbances about which we are unaware are lumped together and called Random or Residual. Hence the errors caused by these happenings are called Random errors.

Characteristics of Instruments:-

(A) Static characteristics

- (i) Accuracy
- (ii) Precision
- (iii) Sensitivity
- (iv) Reproducibility
- (v) Drift
- (vi) Dead zone
- (vii) Dead time
- (viii) Static error
- (ix) Repeatability

(B) Dynamic error

- (i) Speed of response
- (ii) Measuring lag
- (iii) Fidelity
- (iv) Dynamic error

(C) Others

- (i) Reliability
- (ii) Resolution
- (iii) Range & span

1.2 Desirable qualities of measuring Instruments.

(i) Static characteristics:

These characteristics are those that must be considered when the system or instrument is used to measure a condition not varying with time.

Following are the main static characteristics:

(i) Accuracy: — It is defined as the degree of exactness (closeness) of a measurement compared to the expected (desired) value. Thus, accuracy of a measurement means conformity to truth. Ex: — Measured value $\rightarrow A_m$
Actual (true) value $\rightarrow A_t$
 $A_m \uparrow$ (A_m close to A_t)
 $A_m \downarrow$ (A_m far from A_t)

(ii) Precision: —

It is the measure of consistency or repeatability of measurements.

Ex: — $(A_t) - A_{m1}$

$(A_t) - A_{m2}$

$\vdots$

$(A_t) - A_n$

n — times measurement same quantity

$\{ A_{m1} = A_{m2} = \dots = A_{mn} \}$

(iii) Sensitivity: —

The static sensitivity of an instrument or system is the ratio of the magnitude of the output signal or response to the magnitude of input signal or the quantity being measured. Its units are mm/μA, counts/volt etc. depending upon type of input and output.

(iv) Resolution: —

It is the smallest change in the input signal (quantity under measurement) which can be detected by the instrument.

(v) Reproducibility: —

It is the degree of closeness with which a given value may be repeatedly measured. Perfect reproducibility means that the instrument has no drift. It is specified in terms of scale reading over a given period of time.

on the other hand, Repeatability is defined as the variation of the scale reading and is random in nature.

(vi) Drift: — Drift is undesired change in the output-input relation over a period of time. It is a slow variation in the o/p signal of a transducer or measuring system.

which is not due to any change in the input quantity.

(vii) Static error: — It is defined as algebraic difference between the value indicated or conveyed by the output of measuring system and the true value of the quantity presented to the input.

$$\text{i.e. Static error} = \text{Measured value} - \text{True value}$$

The ratio of static error to the true value of the quantity under measurement is called Relative error.

$$\begin{aligned} \text{Relative error} &= \frac{\text{Static error}}{\text{True value}} \\ &= \frac{\text{Measured value} - \text{True value}}{\text{True value}} \end{aligned}$$

(viii) Dead zone: — It is defined as the largest change of input quantity for which there is no output of the instrument.

Dynamic characteristics: —

following are the dynamic characteristics of a measurement system: —

(i) Speed of response: — It is defined as the rapidity with which a measurement system responds to change in measured quantity.

(ii) Measuring lag: — It is ~~defined as~~ the retardation or delay in the response of a measurement system to changes in the measured quantity.

(iii) Fidelity: — it is defined as the degree to which a measurement system indicates changes in the measured quantity without any dynamic error.

(iv) Dynamic: — it is the difference betⁿ the true value of the quantity (under measurement) changing with time and the value indicated by the measurement system if no static error is assumed. it is also called measuring error.